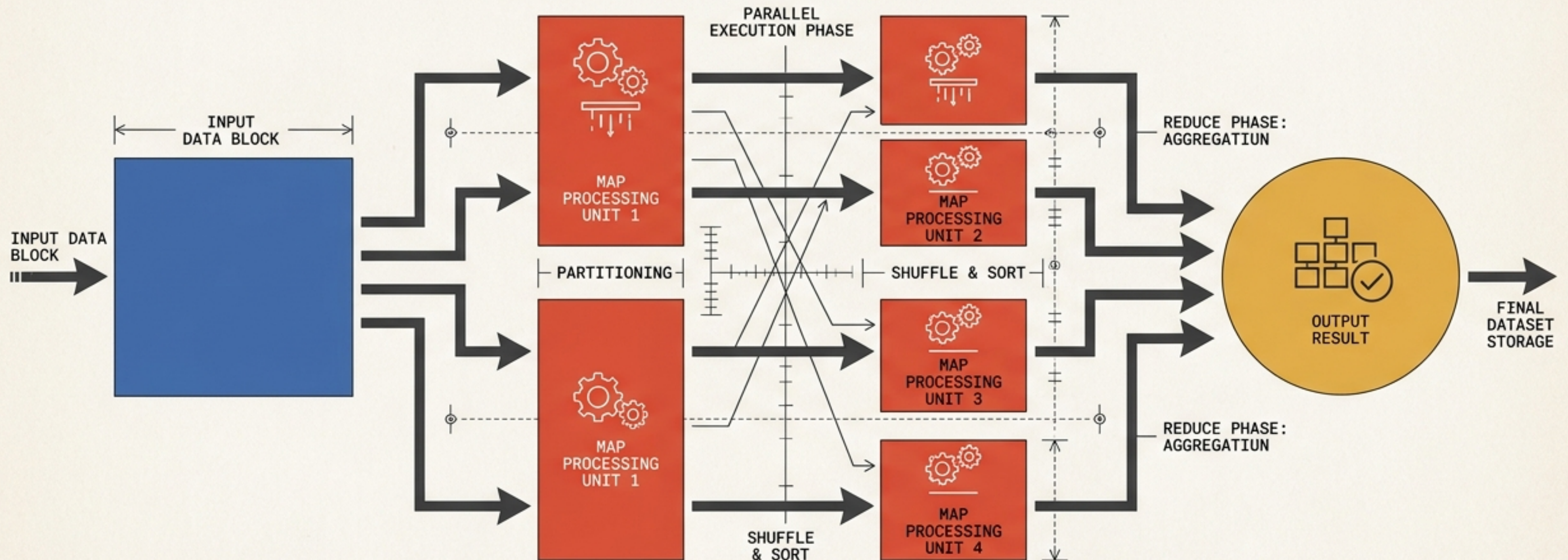


MAPREDUCE ARCHITECTURE

The Distributed Data Factory

Comprehensive Study Notes & Production Cluster Mechanisms

Data Engineering Session 04

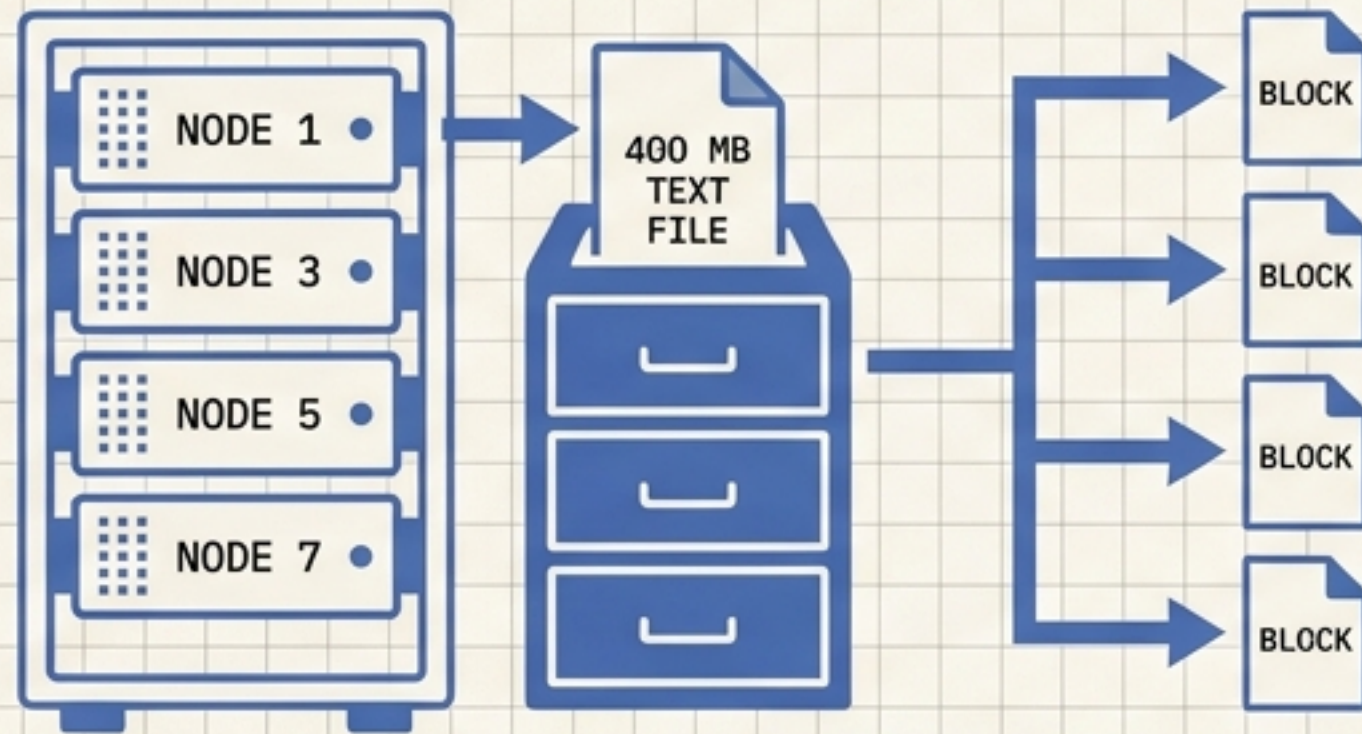


THE HADOOP ECOSYSTEM CORE: STORAGE VS. ANALYSIS

HDFS (Hadoop Distributed File System)

Role: Storage.

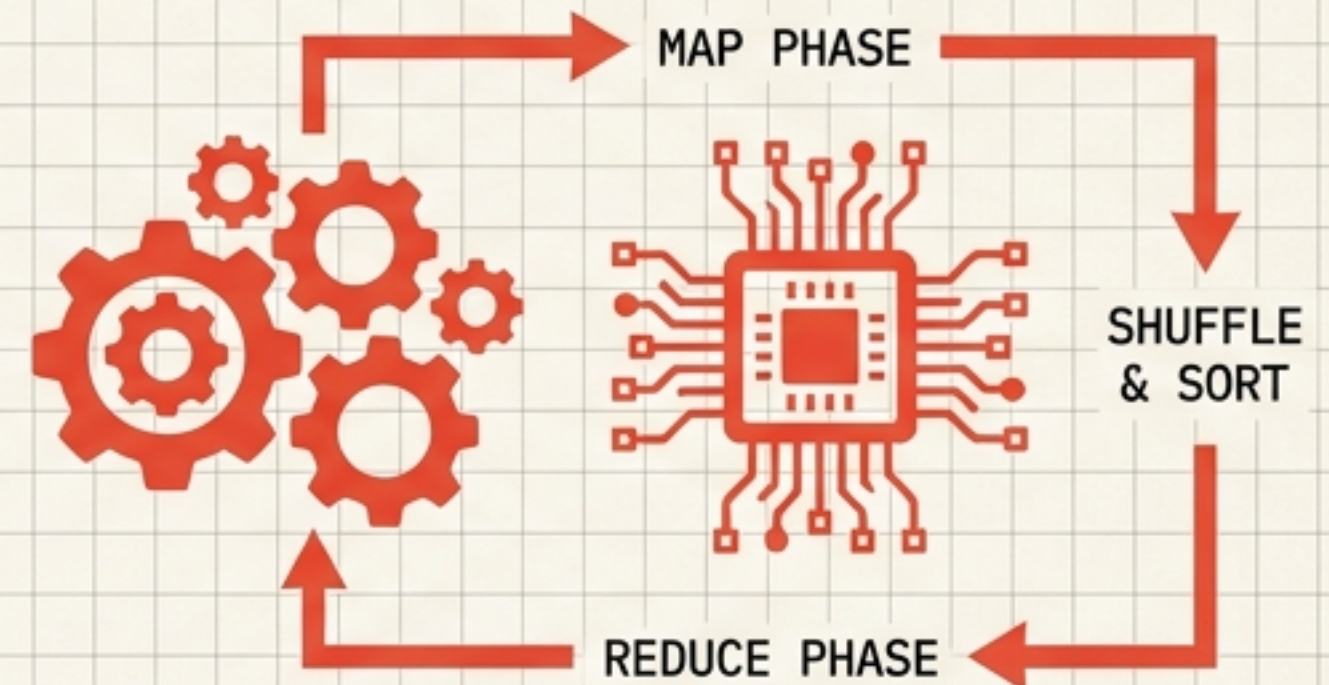
Mechanism: Stores massive datasets (e.g., a 400 MB text file) by splitting them across multiple isolated DataNodes (e.g., Nodes 1, 3, 5, 7).



MapReduce

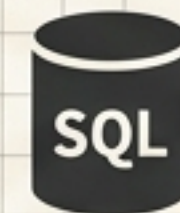
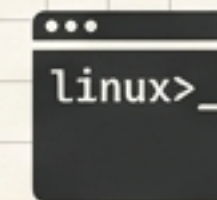
Role: Data Analysis & Processing.

Mechanism: The foundational background compute process that executes automatically when analytical queries are run on distributed data.



THE INVISIBLE ENGINE

MapReduce is the underlying compute mechanism for the Hadoop ecosystem. Whether running Linux commands via Sqoop or executing SQL queries (HiveQL) via Hive, MapReduce is the factory floor performing the heavy lifting behind the scenes.



THE MACRO GOAL: FROM RAW DATA TO AGGREGATED INSIGHTS

BEFORE



Raw Input: Unstructured
rows stored in HDFS.
Format: [ID, Name, Address]

PROCESS

```
SELECT Name,  
COUNT(Name)  
FROM employees  
GROUP BY Name;
```

The Query: Client
requests specific
analytical insights.

AFTER

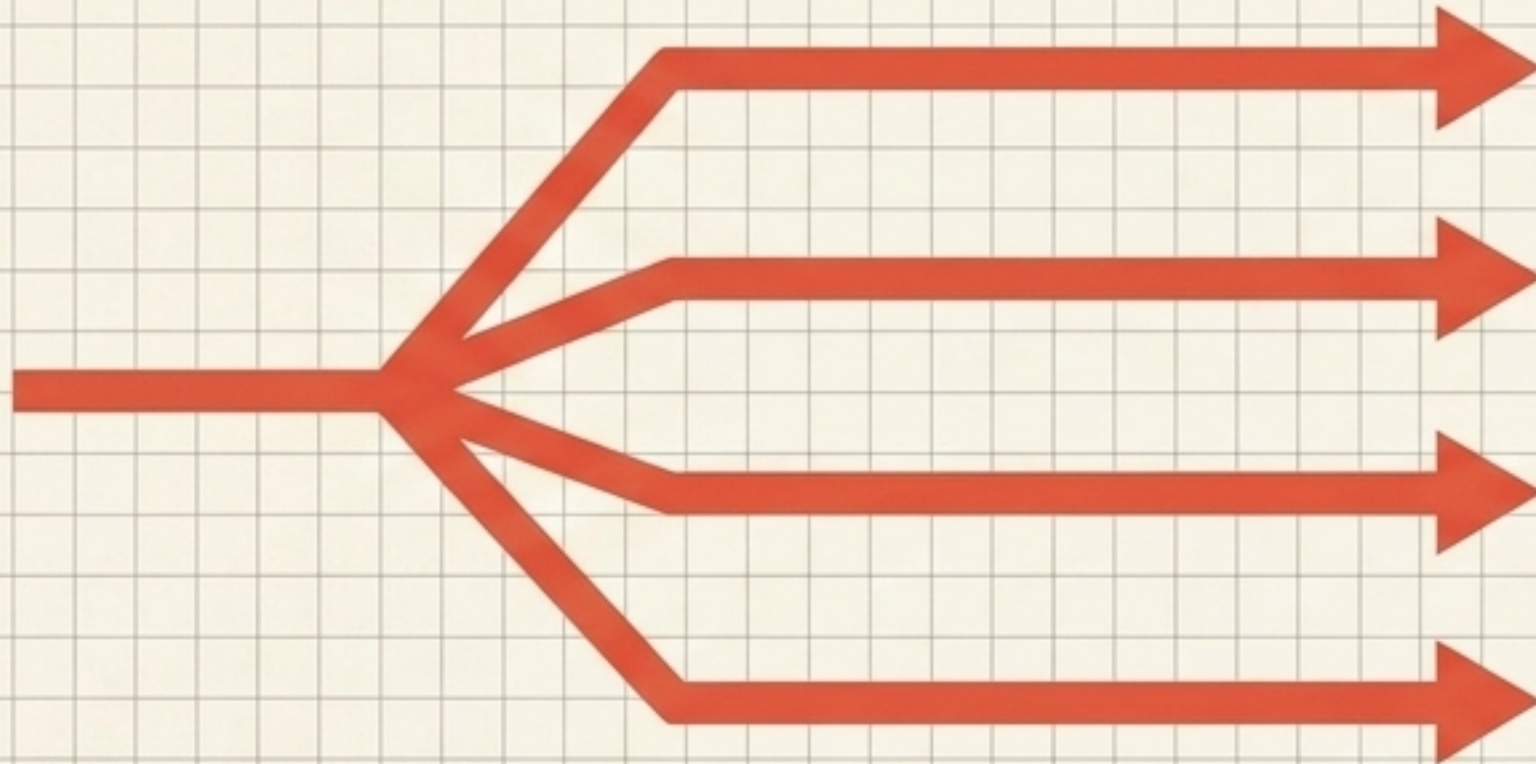
Name	Count
Swap	10
Raj	11
Ravi	8

Structured Output:
Aggregated unique names
and exact frequencies.

The Bridge: To get from the raw distributed HDFS file to the aggregated insights,
the system executes the MapReduce Internal Architecture.

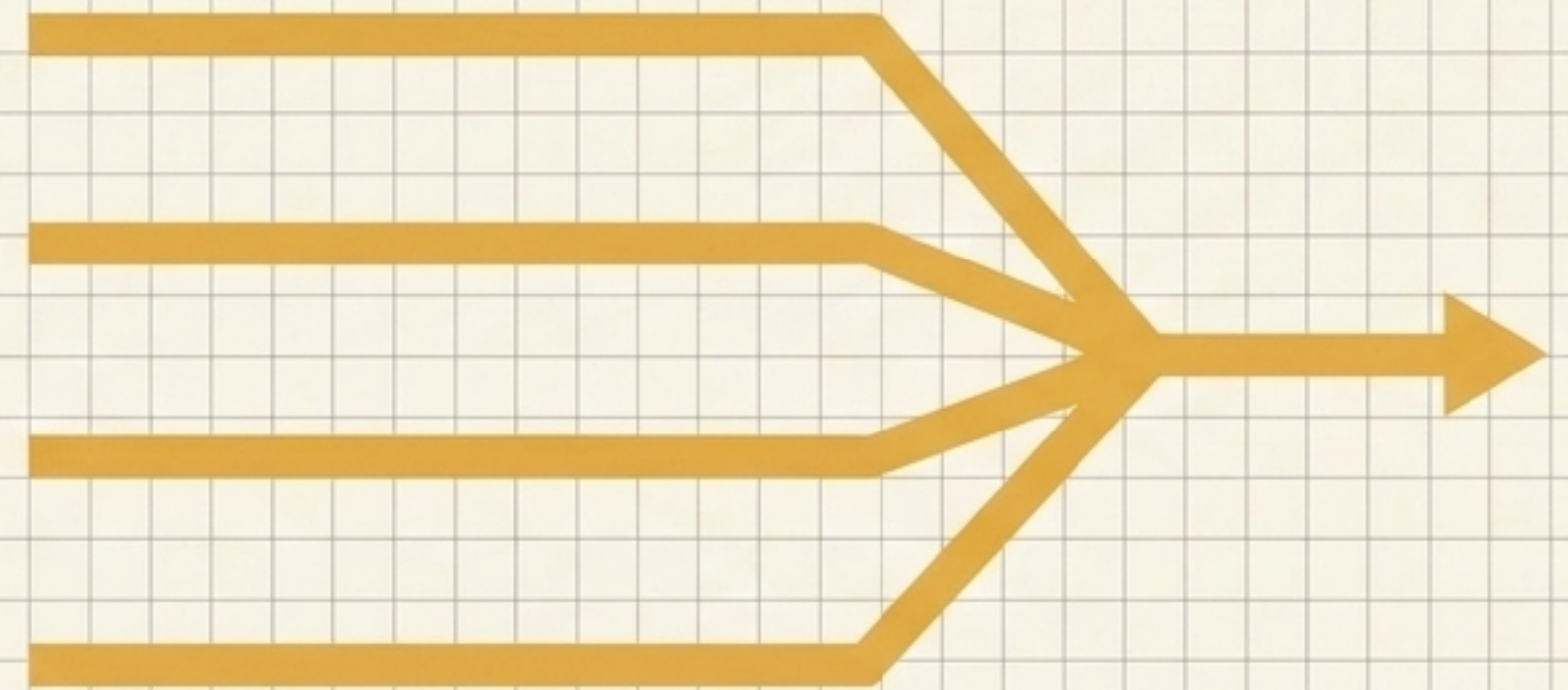
INTERNAL ARCHITECTURE: THE TWO PRODUCTION PHASES

ZONE 1: THE MAPPING PHASE



- Function: Data ingestion, parallel reading, and initial structural transformation.
- Key Components: Input Splits, Record Readers, Map Blocks.

ZONE 2: THE REDUCING PHASE



- Function: Local aggregation, global consolidation, and final document output.
- Key Components: Combiners, The Reducer, Part File.

Mapping Phase (Step 1): Input Splits & Record Readers

DataNode 1

DataNode 3

DataNode 5

DataNode 7

Input Splits: The 400MB file is distributed as physical chunks (A.txt, B.txt, etc.).
The analytical query travels parallelly to these specific nodes.

Record Reader

Record Reader

Record Reader

Record Reader

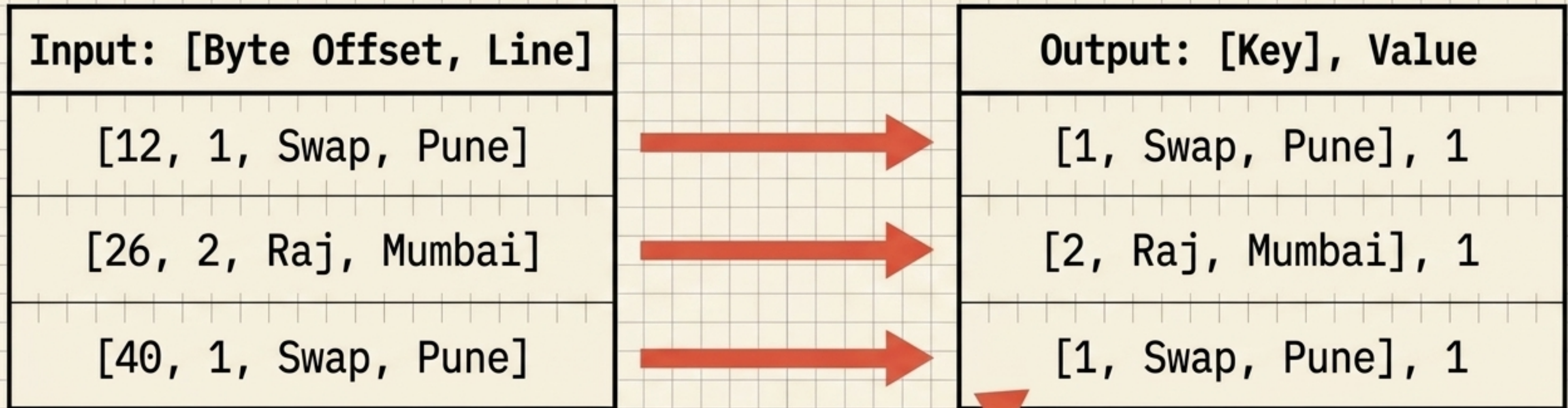
Record Reader: Assigned to each Input Split. Sequentially reads the raw text from the physical data chunk.

OUTPUT FORMAT: [Byte Offset, Entire Line]
EXAMPLE: [0, '1, Swap, Pune']

Byte Offset: The starting character address/index of the line. The first line is always 0.
Entire Line: The actual raw text record.

MAPPING PHASE (STEP 2): THE MAP BLOCK

The Core Task: The Map Block receives the output from the Record Reader. It intentionally discards the Byte Offset and generates static Key-Value pairs based on the analytical query requirements.

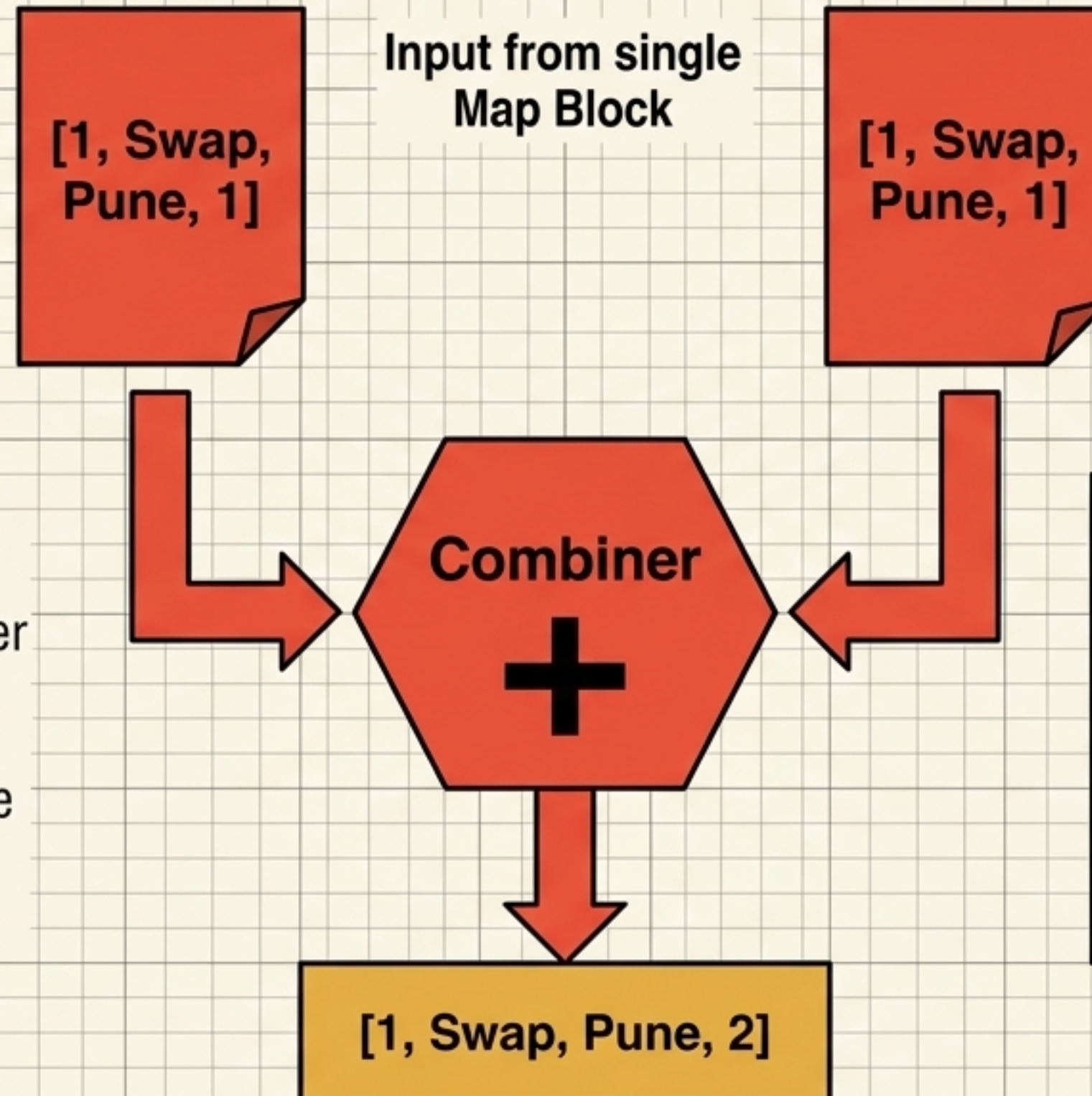


Transformation Logic: The assigned value in this phase is entirely static. Even when a data row repeats exactly, the Map Block blindly assigns a value of 1.

Reducing Phase (Step 1): Local Aggregation via Combiners

The Combiner:

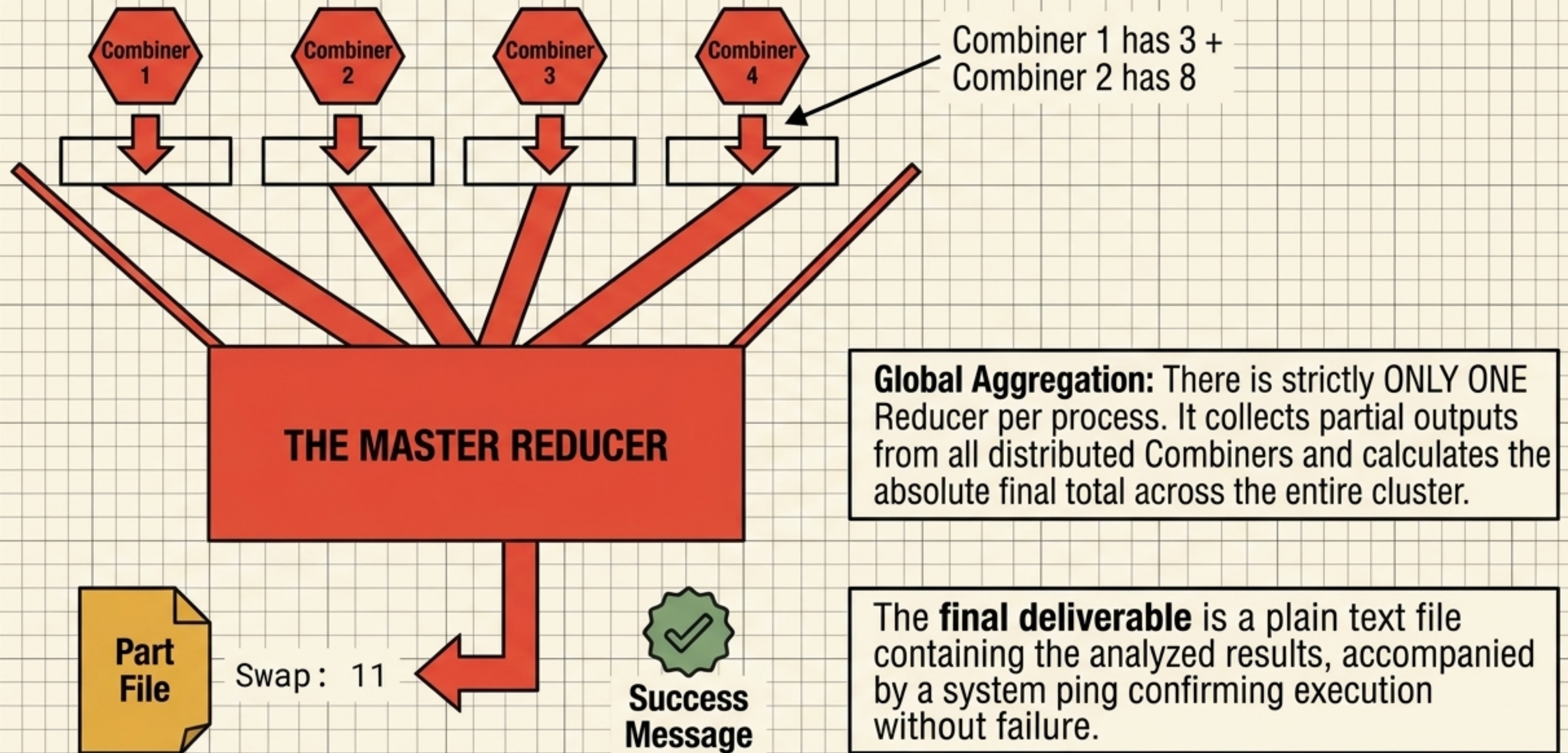
Acts as a localized mini-reducer operating exclusively for its specific Map Block. Its purpose is to drastically reduce network payload before global transmission.



The Golden Rule of Combining:

- Identify repeating keys within the local data pool.
- Keep the key completely constant.
- Mathematically sum the values for every duplicate found.

REDUCING PHASE (STEP 2): GLOBAL OUTPUT & THE PART FILE



Architecture Topology: The Golden Rule

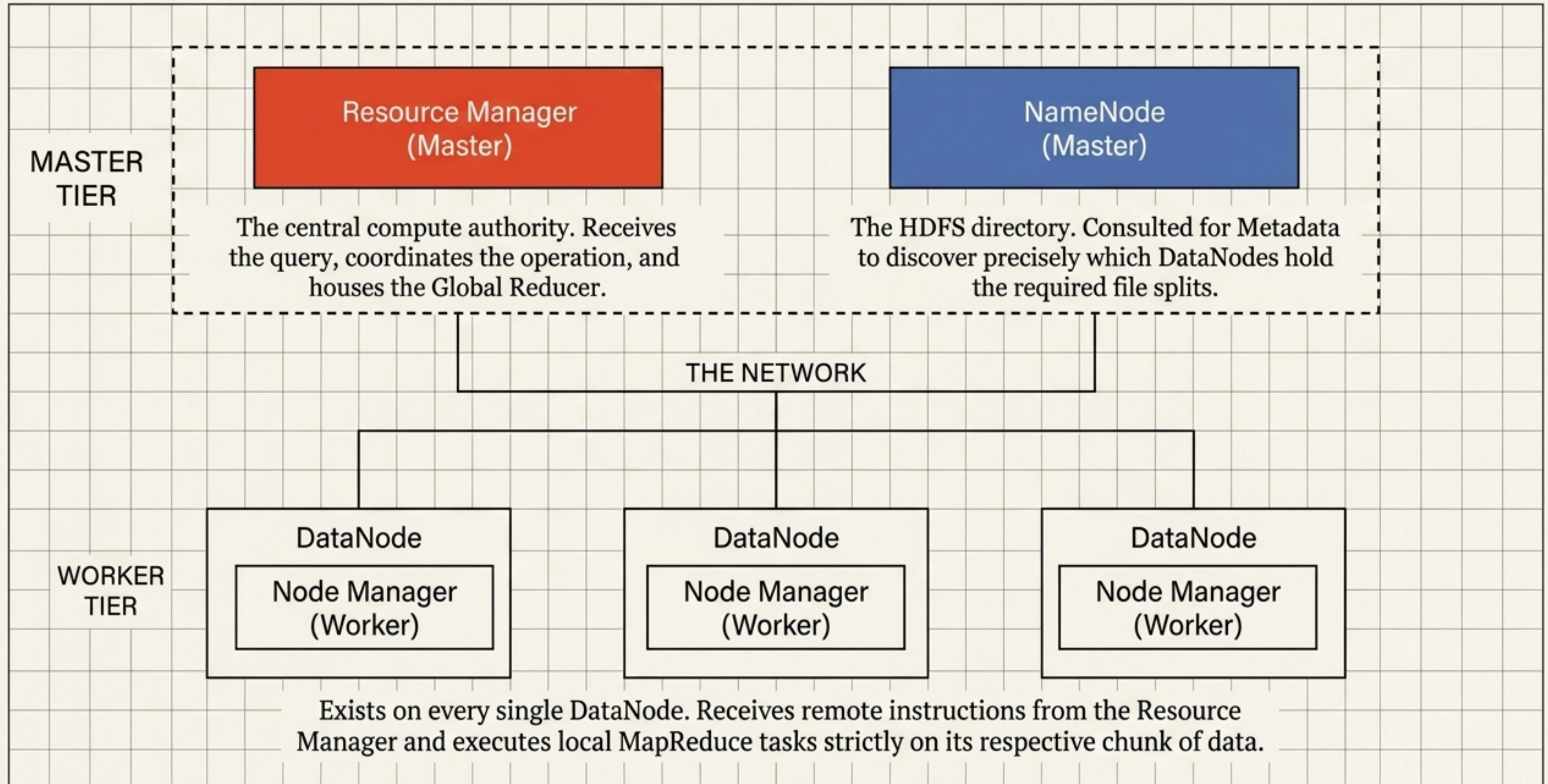
$$\left[\begin{array}{c} \text{Input} \\ \text{Splits} \end{array} \right] = \left[\begin{array}{c} \text{Record} \\ \text{Readers} \end{array} \right] = \left[\begin{array}{c} \text{Map} \\ \text{Blocks} \end{array} \right] = \left[\begin{array}{c} \text{Combiners} \end{array} \right]$$

To understand cluster scaling, remember this strict parallel equivalence. The number of physical data chunks physically stored dictates the precise number of processing components instantiated.

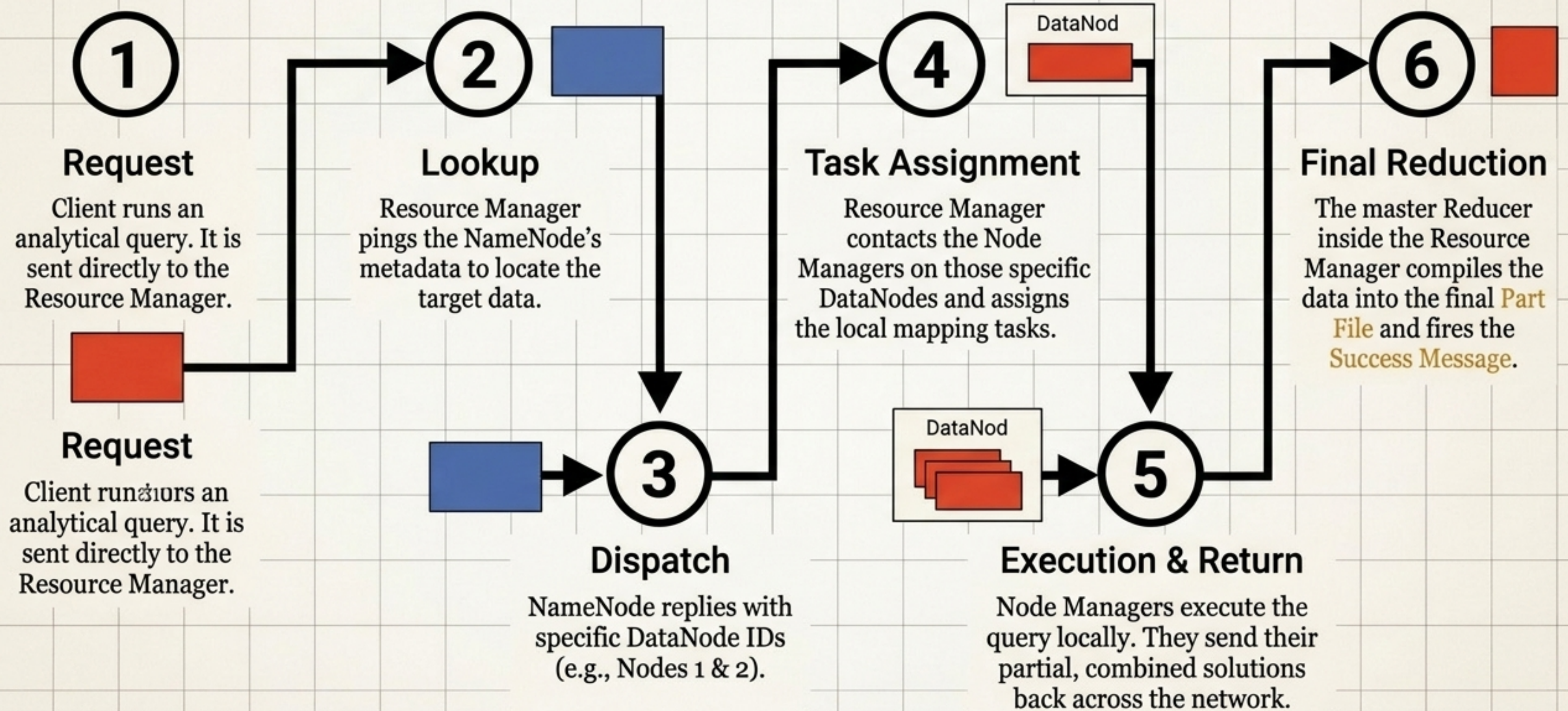
THE EXCEPTION:

There is ALWAYS only 1 Reducer. No matter how many splits or combiners exist, they must all funnel into a single Reducer to ensure a single, consolidated output Part File.

Zooming Out: Production Cluster Architecture



The Query Lifecycle in Production



MapReduce: Quick-Reference Study Guide

Record Reader Output

Format: [Byte Offset, Entire Line]

Meaning:

Byte Offset is the numeric character address.
Entire Line is the actual unstructured data record extracted from the split.

Map Block Task

Task: Converts raw Record Reader output into analyzable [Key, Value] pairs.

Detail: The structural Value is typically statically assigned as '1', discarding the byte offset completely.

Combiner Purpose

Purpose: Acts as a localized **mini-reducer** for a specific map block.

Benefit: Pre-aggregates duplicate keys locally, drastically **minimizing the data payload** that must be sent across the network.

The HDFS Connection

Relationship: **HDFS** stores the massive files in distributed physical chunks.

Compute Engine: **MapReduce** is the backend **compute engine** that travels directly TO those physical chunks to analyze the data locally, rather than moving the data to the compute.